

NE 511: Multi-Physics of Nuclear Reactors

UNIT 2: Short-Time Multi-Physics Phenomena in Reactor Core

Lecture 5 : *PRELIMINARY FORMULATION OF THE POINT KINETICS:*

PROMPT NEUTRON BALANCE EQUATION

INTUITIVE POINT KINETICS EQUATION

ONE-GROUP POINT KINETICS EQUATION

DEGREE OF FREEDOM IN NEUTRON BALANCE EQUATION

Neutron balance equation in diffusion theory approximation:

!! *Five independent variables (Cartesian geometry):*

1. *Time*
2. *Energy*
3. *Space: x-direction*
4. *Space: y-direction*
5. *Space: z-direction*

Neutron balance equation in transport theory approximation:

!! *Seven independent variables (Cartesian geometry):*

1. *Time*
2. *Energy*
3. *Space: x-direction*
4. *Space: y-direction*
5. *Space: z-direction*
6. *Angle: polar*
7. *Angle: azimuthal*

“POINT KINETICS” ...?

What is the optimum degree of freedom in neutron balance equation?

It is case-dependent.

- For many applications it is neither necessary nor feasible to solve a complicated set of equations including the space-dependent set of precursor equations.

- Degree of freedom in the Point Kinetics Equations:

The equations can be condensed into purely time-dependent so-called "point kinetics" equations.

The Prompt Neutron Balance Equation
&
The Intuitive Point Kinetics Equations

THE PROMPT NEUTRON BALANCE EQUATION

Let's consider:

- An off-critical reactor*
- No independent source*
- All delayed neutrons are assumed to be promptly emitted*
- Single homogeneous composition*

Off-criticality: loss $\neq$ production

→ The off-balance leads to a time variation proportional to $n(t)$:

$$\dot{n}(t) = \frac{dn(t)}{dt} = \nu \Sigma_f \hat{\Phi}(t) - (\Sigma_a + DB^2) \hat{\Phi}(t)$$

THE PROMPT NEUTRON BALANCE EQUATION

One-group (energy) model → the total integrated flux and the number of neutrons are related by an average neutron velocity:

$$\hat{\Phi} = \bar{v} n \qquad \frac{1}{\bar{v}} \frac{d\hat{\Phi}(t)}{dt} = (v \Sigma_f - \Sigma_a - DB^2) \hat{\Phi}$$

Prompt Neutron Balance Equation in terms of reactivity:

$$k = \frac{v \Sigma_f}{\Sigma_a + DB^2} \qquad \longrightarrow \qquad \frac{1}{\bar{v} v \Sigma_f} \frac{d\hat{\Phi}}{dt} = \rho \hat{\Phi}$$
$$\rho = \frac{k - 1}{k}$$

THE PROMPT NEUTRON BALANCE EQUATION

Some Discussions

1. The average traveling distance between birth and absorption or leakage ("death") of a neutron is $\frac{1}{\Sigma_a + DB^2}$ and thus:

$$\frac{1}{\bar{v}} \frac{1}{(\Sigma_a + DB^2)} = \ell$$

2. For an infinite system (no leakage - $B^2=0$): $\ell_\infty = \frac{1}{\bar{v}} \frac{1}{\Sigma_a}$

3. Relation between the lifetime in a finite system and the lifetime in an infinite system:

$$k = \frac{\nu \Sigma_f}{\Sigma_a} \frac{1}{1 + L^2 B^2} = \frac{k_\infty}{1 + L^2 B^2} \quad \ell = \frac{\ell_\infty}{1 + L^2 B^2}$$

where L is the one-group diffusion length, $L = \sqrt{D/\Sigma_a}$

4. The one-group generation time is not directly related to the size of the system, only to the one-group lifetime contains DB^2

THE PROMPT NEUTRON BALANCE EQUATION

Some Discussions

5. The lifetime and the generation time are directly proportional to each other, and the proportionality constant is just the multiplication constant k :

$$\ell = k\Lambda.$$

This equation shows that the average lifetime is smaller than the average neutron generation time if the reactor is subcritical, and it is larger than if the reactor is supercritical.

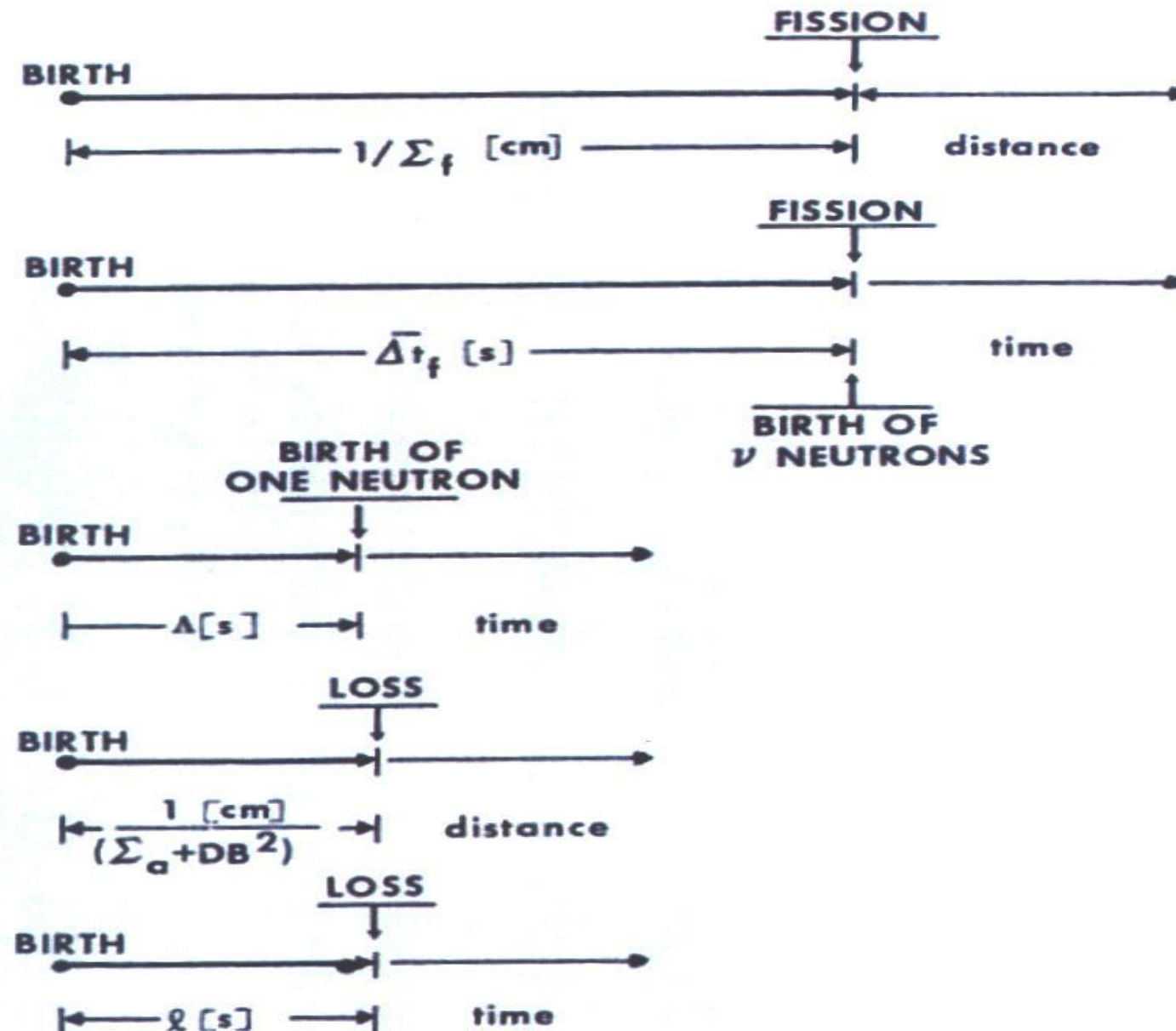
6. By introducing either the generation time or the lifetime into the prompt neutron kinetics equation, we obtain:

$$\frac{d\hat{\Phi}}{dt} = \frac{\rho}{\Lambda} \hat{\Phi} \qquad \frac{d\hat{\Phi}}{dt} = \frac{\Delta k}{\ell} \hat{\Phi}$$

THE PROMPT NEUTRON BALANCE EQUATION

Some Discussions

Illustration of the interpretation of neutron generation time and lifetime



THE INTUITIVE POINT KINETICS EQUATIONS

Let's consider:

- An off-critical reactor
- No independent source
- Delayed neutrons are accounted
- Single homogeneous composition

To account for this split:

$$\frac{1}{\bar{v}} \frac{d\hat{\Phi}}{dt} = (\nu_p \Sigma_f - \Sigma_a - DB^2) \hat{\Phi} + \sum_k \lambda_k \hat{C}_k$$

OR:

$$\frac{1}{\bar{v} \nu \Sigma_f} \frac{d\hat{\Phi}}{dt} = \left[\frac{\nu \Sigma_f - (\Sigma_a + DB^2)}{\nu \Sigma_f} - \frac{\nu_d \Sigma_f}{\nu \Sigma_f} \right] \hat{\Phi} + \frac{1}{\nu \Sigma_f} \sum_k \lambda_k \hat{C}_k$$

$$\Lambda \frac{d\hat{\Phi}}{dt} = (\rho - \beta) \hat{\Phi} + \frac{1}{\nu \Sigma_f} \sum_k \lambda_k \hat{C}_k$$

THE INTUITIVE POINT KINETICS EQUATIONS

Derivations

The reactivity ρ is given by its one-group approximation.

The delayed neutron fraction, β , is
$$\beta = \frac{\nu_d \Sigma_f}{\nu \Sigma_f}$$

The "effective delayed neutron fraction," $\beta_{eff} = \beta$, can also be expressed in terms of contributions of the six delay groups:

$$\beta = \sum_{k=1}^6 \beta_k \quad \text{with} \quad \beta_k = \frac{\nu_{dk} \Sigma_f}{\nu \Sigma_f}$$

THE INTUITIVE POINT KINETICS EQUATIONS

Derivations

Dividing $\Lambda \frac{d\hat{\Phi}}{dt} = (\rho - \beta)\hat{\Phi} + \frac{1}{\nu \Sigma_f} \sum_k \lambda_k \hat{C}_k$ by Λ gives the typical form of the kinetics equation:

$$\frac{d\hat{\Phi}}{dt} = \frac{\rho - \beta}{\Lambda} \hat{\Phi} + \frac{1}{\Lambda \nu \Sigma_f} \sum_k \lambda_k \hat{C}_k$$

This equation must be completed by equations describing the balance between production and decay of precursors:

$$\frac{d\hat{C}_k}{dt} = -\lambda_k \hat{C}_k + \nu_{dk} \Sigma_f \hat{\Phi}, \quad k = 1, \dots, 6$$

These equations represent a set of seven differential equations for the seven unknown functions $\hat{\Phi}(t)$ and $\hat{C}_k(t)$ with $k = 1, \dots, 6$, and are termed the intuitive point kinetics equations.

The One-Group Point Kinetics Equations

ONE-GROUP POINT KINETICS EQUATIONS

Some Discussions

If the stationary neutron balance is perturbed at some area in space, the result will be a time dependence of the entire neutron flux.

This is described by the time-dependent diffusion equation:

$$\frac{1}{\nu} \frac{d\Phi(r, E, t)}{dt} = (F_p - M)\Phi(r, E, t) + S_d(r, E, t) + S(r, E, t)$$

This equation is the space-dependent analog to the intuitive point kinetics equations written in an operator form.

ONE-GROUP POINT KINETICS EQUATIONS

Some Discussions

The operator $\mathbf{F}_p$ describes the production of prompt neutrons only,

$$F_p \Phi = \sum_i \chi_{pi}(E) \int_{E'} \nu_{pi}(E') \Sigma_{fi}(r, E', t) \Phi(r, E', t) dE'$$

The operator $\mathbf{M}$ is defined in the same way as it is for static problems

$$\begin{aligned} M\Phi = & -\nabla \cdot D(r, E, t) \nabla \Phi(r, E, t) + \Sigma_t(r, E, t) \Phi(r, E, t) \\ & - \int_{E'} \Sigma_s(r, E' \rightarrow E, t) \Phi(r, E', t) dE' \end{aligned}$$

and

$$S_d(r, E, t) = \sum_k \lambda_k C_k(r, t) \chi_{dk}(E)$$

Some Discussions

1. The isotope dependence of χ_{pi} is for the most part neglected:
 - *The prompt neutron emission spectra of ^{238}U and of the four Pu isotopes that occur together in fast reactors are not significantly different.*
 - *$\chi_{pi}(E)$ of ^{235}U differs somewhat from the emission spectra of Pu isotopes.*

For the prompt fission source, disregarding the isotope dependence χ_p of gives:

$$F_p \Phi = \chi_p(E) \int_{E'} \nu_p(E') \Sigma_f(r, E', t) \Phi(r, E', t) dE'$$

2. The macroscopic fission cross-section is given by:

$$\nu_p \Sigma_f(r, E, t) = \sum_i \nu_{pi}(E) \Sigma_{fi}(r, E, t)$$

ONE-GROUP POINT KINETICS EQUATIONS

Some Discussions

3. The balance equations for the precursors must be introduced for completeness.
4. In most reactor problems, the physical transport of precursors can be neglected.

No need of a precursor transport term in the balance equations, i.e., the precursors can be assumed to decay at the location where they are produced -"stationary fuel".

The balance equation for the precursor concentrations produced by all isotopes is given by:

$$\frac{dC_k(r,t)}{dt} = -\lambda_k C_k(r,t) + \int_0^\infty \nu_{dk} \Sigma_f(r, E', t) \Phi(r, E', t) dE'$$

$$\nu_{dk} \Sigma_f(r, E, t) = \sum_i \nu_{dki} \Sigma_{fi}(r, E, t)$$

ONE-GROUP POINT KINETICS EQUATIONS

Some Discussions

The generation time Λ now depends on the average inverse velocity, rather than the inverse average value:

$$\Lambda = \frac{1}{v \Sigma_f} \left(\frac{\bar{1}}{v} \right)$$

To simplify the notation, the following relative sources are introduced:

Relative neutron source of delayed group k ($\zeta_k(t)$ is called the “reduced precursors” of delayed group k):

$$\lambda_k \zeta_k(t) = \lambda_k \frac{\hat{C}_k(t)}{\hat{S}_{f0}}$$

Relative independent source: $s(t) = \frac{\hat{S}(t)}{\hat{S}_{f0}}$

ONE-GROUP POINT KINETICS EQUATIONS

Some Discussions

After integrating the precursor balance equation with respect to space and dividing by $\hat{S}_{f0}$ one obtains a differential equation for the reduced precursors:

$$\frac{d\zeta_k(t)}{dt} = -\lambda_k \zeta_k(t) + \beta_k p(t)$$

This yields the improved definition of β_k , i.e.,

$$\beta_k = \frac{\int_V \int_0^\infty v_{dk} \Sigma_f(r, E) \psi(r, E) dE dV}{\int_V \int_0^\infty v \Sigma_f(r, E) \psi(r, E) dE dV}$$

ONE-GROUP POINT KINETICS EQUATIONS

Some Discussions

The [approximate point kinetics equations](#) are then given by

$$\frac{dp}{dt} = \frac{\rho - \beta}{\Lambda} p + \frac{1}{\Lambda} \sum_k \lambda_k \zeta_k + \frac{1}{\Lambda} s(t)$$

$$\frac{d\zeta_k}{dt} = -\lambda_k \zeta_k + \beta_k p(t)$$

The definitions of all quantities are expressed in terms of one-group model.

ONE-GROUP POINT KINETICS EQUATIONS

Some Discussions

Commonly, the $1/\Lambda$ factors in front of the sources are combined with these quantities:

$$c_k(t) = \frac{1}{\Lambda} \zeta_k(t) \qquad s_c(t) = \frac{1}{\Lambda} s(t)$$

This results in the more familiar form of the point kinetics equations:

$$\frac{dp}{dt} = \frac{\rho - \beta}{\Lambda} p + \sum_k \lambda_k c_k + s_c$$
$$\frac{dc_k}{dt} = -\lambda_k c_k + \frac{\beta_k}{\Lambda} p$$

The Exact Point Kinetics Equations

THE EXACT PKE...?

“Exact” in this context means that a given time-, space-, and energy-dependent neutronics model is condensed or lumped into the form of the point kinetics equations without applying simplifying approximations.

The conceptual starting point is the factorization of the neutron flux:

$$\phi(\mathbf{r}, E, t) = p(t) \cdot \psi(\mathbf{r}, E, t)$$

Flux Factorization and Weighting Functions

THE 1ST GENERALIZATION

Applicability of the neutron flux factorization...?

Flux amplitude changes + Comparatively small time variations of the shape function.

A second equation is required to make the factorization unique.

A specific integral of the shape function ψ is introduced, which is held constant in time, in order to obtain the desired form of the point kinetics equations.

Allowing the shape function to depend on time the first generalization compared to the use of time-independent shape in the preliminary one-group PKE.

THE 2ND GENERALIZATION

Freedom of choice by **applying an arbitrary weighting function**:

The neutronics equation is multiplied by a weight function, $w(r,E)$, prior to integration with respect to space and energy.

Introducing the flux factorization into the left side of time-dependent diffusion equation yields:

$$\begin{aligned} \int_V \int_0^\infty \frac{w(\mathbf{r}, E)}{\nu(E)} \frac{\partial \phi(\mathbf{r}, E, t)}{\partial t} dE dV &= \frac{dp(t)}{dt} \int_V \int_0^\infty \frac{w(\mathbf{r}, E) \psi(\mathbf{r}, E, t)}{\nu(E)} dE dV \\ &+ p(t) \frac{d}{dt} \int_V \int_0^\infty \frac{w(\mathbf{r}, E) \psi(\mathbf{r}, E, t)}{\nu(E)} dE dV \end{aligned}$$

THE 3RD GENERALIZATION

Constrain the time variation of the shape function and thus make the factorization unique:

$$\int_V \int_0^\infty \frac{w(\mathbf{r}, E) \psi(\mathbf{r}, E, t)}{\nu(E)} dE dV = \text{constant}$$

FLUX FACTORIZATION AND WEIGHTING FUNCTIONS

1. Factorizing the neutron flux, applying an arbitrary weighting function, and constraining the variation of the flux shape in this way do NOT introduce an approximation.
2. The factorization introduces a new degree of freedom (two instead of one function), which needs to be eliminated again by a constraint condition.
3. The weight function only influences the precise split of the flux into amplitude and shape function as well as the definitions of all other quantities appearing in the point kinetics equations.

The end result for the flux is unique despite the dependence of the magnitude of both factors on the weighting function.

$$\phi(\mathbf{r}, E, t) = p^w(t) \cdot \psi^w(\mathbf{r}, E, t)$$

SELECTION OF THE WEIGHT FUNCTION...?

1. Use of an approximate shape functions, for example $\psi(\mathbf{r}, E, t) \cong \phi_0(\mathbf{r}, t)$.
2. It is obtained by extrapolation of shape functions at earlier times.

Since the shape function is not known precisely, the resultant flux and particularly the flux amplitude will only be approximate solutions.

→ Both will depend on the choice of the weight function:

→ It is thus advantageous to choose the weight function in such a way that it reduces the error resulting from inaccuracies in the shape function.

SELECTION OF THE WEIGHT FUNCTION...?

Since the solution of the point kinetics equation is particularly sensitive to an error in the reactivity, a weight function should be chosen such that it reduces the effect of the shape function inaccuracies on the reactivity.

With **the initial adjoint flux** as a weighting function,

$$w(\mathbf{r}, E) = \phi_0^*(\mathbf{r}, E)$$

the constraint condition for the shape function is given by

$$\int_V \int_0^\infty \frac{\phi_0^*(\mathbf{r}, E) \psi(\mathbf{r}, E, t)}{\nu(E)} dE dV = K_0$$

with K_0 being a constant

EXACT POINT KINETICS EQUATIONS

Flux amplitude function:

$$\dot{p}(t) = \frac{\rho(t) - \beta(t)}{\Lambda(t)} p(t) + \frac{1}{\Lambda_0} \sum_k \lambda_k \zeta_k(t)$$

Exact balance equation for the reduced precursors:

$$\dot{\zeta}_k(t) = -\lambda_k \zeta_k(t) + \frac{F(t)}{F_0} \beta_k(t) p(t)$$

$$\beta_k(t) = \frac{1}{F(t)} (\Phi_0^*, \mathbf{F}_{dk} \Psi)$$

$$\zeta_k(t) = \frac{1}{F_0} (\Phi_0^*, \mathcal{X}_{dk} C_k)$$

Next Class *Reactivity in the Exact PKEs*
Effective Delayed Neutron Fractions
Point Reactor Model